

# OLIST E-COMMERCE DATA ANALYSIS

Brazil Marketplace Insights (2016–2018)

Tools: Python | SQL | Pandas | Power BI/Tableau



# PROJECT OVERVIEW

## Objective

Analyze Olist marketplace performance across Brazil's e-commerce ecosystem. Understand patterns in customer behavior, seller dynamics, and regional distribution to derive actionable business insights from 2016-2018 data.

## Key Study Areas

- Sales Performance
- Customer Behavior
- Seller Analysis
- Product Categories
- Geographic Distribution

## Dataset Used

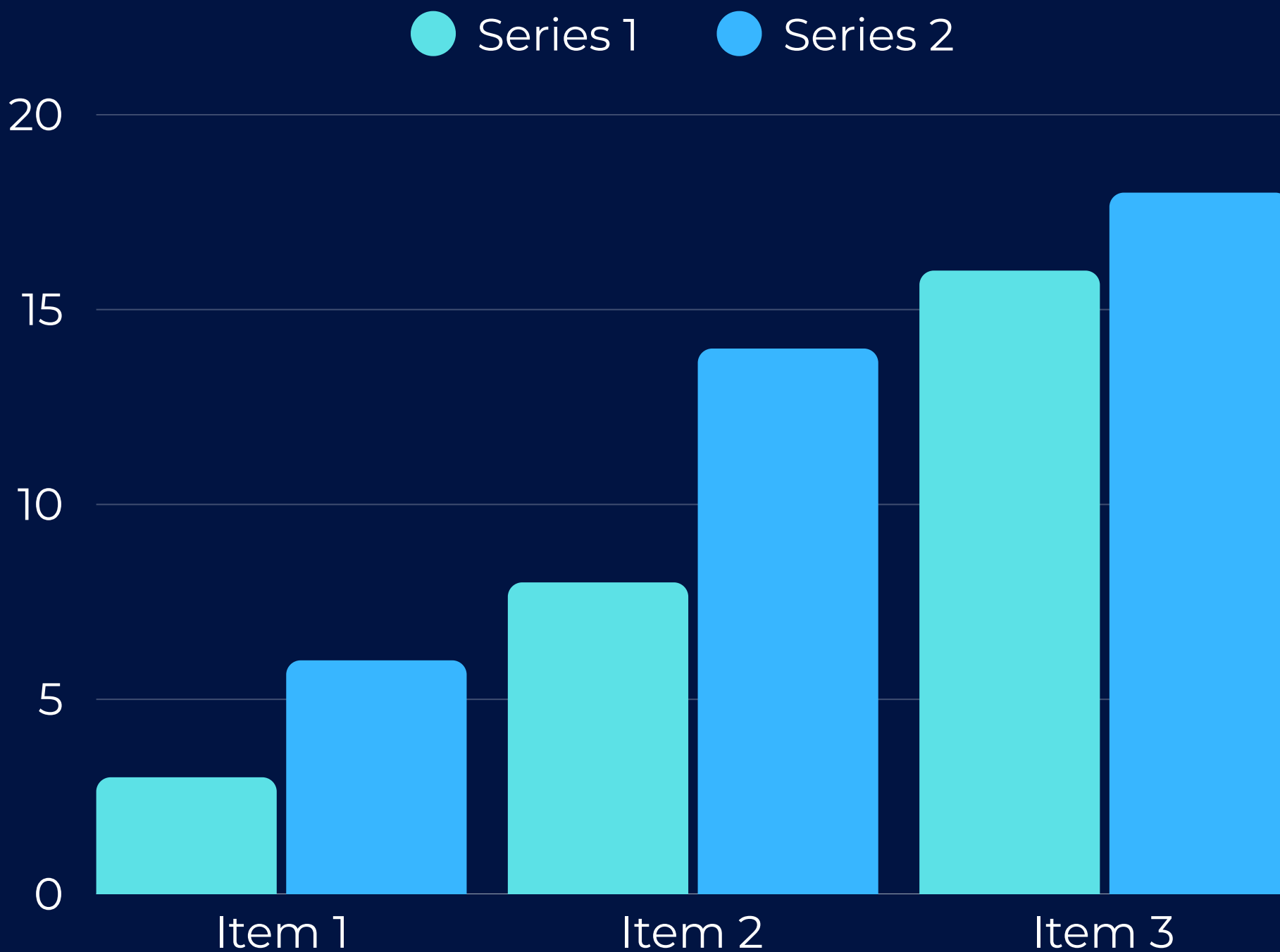
- Orders & Transactions
- Customer Demographics
- Seller Information
- Product Catalog
- Payment Records
- Geolocation Data

## Data Scope

- 2016-2018 timeframe
- Brazilian marketplace focus
- Multiple data tables joined
- Rich transactional detail
- Geographic coordinates included

# OVERALL REVENUE INSIGHTS

Key Metrics: Total Revenue, Average Order Value (AOV), Total Orders, and Freight Revenue form the foundation of Olist's marketplace performance. Analysis reveals that revenue is driven mainly by a few dominant categories and top-performing sellers, following a classic Pareto distribution where roughly 20% of sellers generate 80% of revenue.



# TOP REVENUE CATEGORIES

## Health & Beauty

Highest revenue category in the marketplace. Repeat purchases from loyal customers drive consistent growth. Large and expanding customer base with strong retention rates.

## Bed, Bath & Table

Essential necessity products with consistent demand. Very high order volume due to regular household replenishment needs. Stable revenue stream year-round.

## Watches & Gifts

High-value purchases with premium price points. Seasonal demand during holidays and special occasions significantly boosts quarterly revenue performance.

## Key Takeaway

Revenue concentration follows Pareto distribution —top 3 categories generate majority of marketplace revenue. Focus resources on high-performers.



# HIGHEST AOV CATEGORIES

## Computers

Highest Average Order Value across all categories. Expensive products like laptops, desktops, and accessories significantly increase transaction value per order, making this a premium revenue segment.

## Appliances Categories

High-value household appliances drive substantial transaction values. Large appliances and electronics contribute to elevated AOV, representing key opportunities for marketplace growth.



# CUSTOMER ANALYSIS



Customer Distribution: São Paulo leads with 42% of total customers, followed by Rio de Janeiro (13%) and Minas Gerais (12%). Southeast Brazil dominates marketplace activity.

Purchase Patterns: Average customer places 1.03 orders with AOV of R\$154. Peak ordering occurs on weekdays, especially Monday-Wednesday afternoons.

Repeat Customer Rate: Only 3% of customers made repeat purchases, indicating opportunity for retention strategies and loyalty programs.

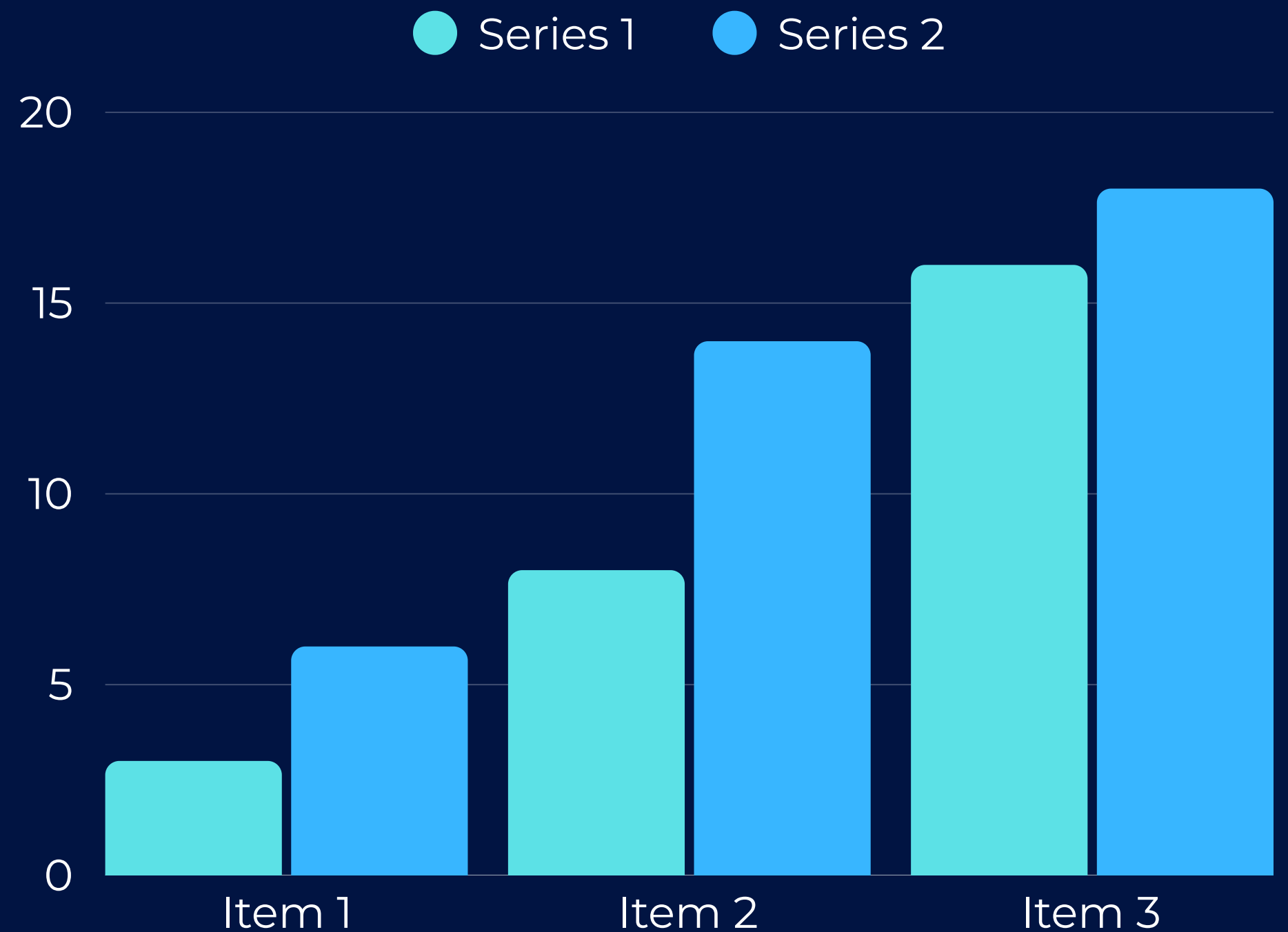
# SELLER PERFORMANCE ANALYSIS

Top performing sellers drive the majority of Olist's marketplace revenue, following a classic Pareto distribution where approximately 20% of sellers generate 80% of total sales. Analysis reveals strong seller-revenue correlation, with high-performing sellers maintaining better ratings, faster delivery times, and broader product catalogs. Seller concentration remains a key marketplace dynamic to monitor for sustainable growth.



# GEOGRAPHIC INSIGHTS

Regional sales analysis reveals significant concentration in Southeast Brazil. São Paulo leads with the highest order volume and revenue contribution, followed by Rio de Janeiro and Minas Gerais. Delivery patterns show urban centers receive faster shipping times, while remote regions face longer transit periods. The geographic distribution follows population density, with coastal states generating 78% of total marketplace revenue.





# KEY FINDINGS & RECOMMENDATIONS

Strategic Recommendations for Olist Marketplace Growth:

1. Focus on Top Categories: Prioritize health\_beauty, watches\_gifts, and bed\_bath\_table categories which drive the majority of revenue. Invest in inventory management and marketing for these high-performing segments.
2. Optimize Seller Network: Identify and support top-performing sellers who contribute disproportionately to revenue. Implement seller training programs to improve overall marketplace quality.
3. Target High-Value Regions: Concentrate expansion efforts in São Paulo, Rio de Janeiro, and Minas Gerais where customer density and purchasing power are highest. Tailor logistics and delivery infrastructure accordingly.



# THANK YOU!



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